MAC21 — Every 7-Marker, Both Sides

All 35 problem types that carry the 7-mark parts, across **both** options in every unit — Q1 and Q2, Q3 and Q4, Q5 and Q6, Q7 and Q8, Q9 and Q10. Nothing here depends on which question you end up answering.

How each entry is laid out: METHOD the recipe, in steps · WORKED one full example, taken from a real paper wherever one exists · TRAP what actually loses marks.
Every numerical answer below has been computed and verified, not estimated. Where a scan was unreadable I say so rather than guess.

UNIT I — Q1 and Q2

1 Newton-Raphson — system of two non-linear equations 4× Q1(c)

METHOD

For $f(x,y)=0$, $g(x,y)=0$: at each step solve the linear system $J\cdot[\Delta x,\ \Delta y]^T = -[f,\ g]^T$ where $J = [[f_x,\ f_y],[g_x,\ g_y]]$, then update $x_{-1} = x + \Delta x$, $y_{-1} = y + \Delta y$. Evaluate J at the *current* point every iteration.

WORKED — M23 1(C) / M24 1(C) / MAM24 1(B)

Solve $x^2+y^2=x$, $x^2-y^2=y$ from $(0.8,\ 0.4)$, two iterations.

Write $f = x^2+y^2-x$, $g = x^2-y^2-y$. Then $J = [[2x-1,\ 2y],[2x,\ -2y-1]]$.

Iter	f	g	Δx	Δy	x	y
start	—	—	—	—	0.800000	0.400000
1	0.0000	+0.0800	−0.027119	+0.020339	0.772881	0.420339
2	+0.0011	+0.0003	−0.001035	−0.000695	**0.771846**	**0.419644**

TRAP

At the start point $f = 0$ exactly — students assume they have made an error and stop. They have not: $g \neq 0$, so the iteration still moves. Also, P26 writes the same system as $x = x^2+y^2$, $y = x^2-y^2$ — identical, just rearranged.

2 Newton-Raphson — single equation 3× Q2(b)/(c)

METHOD

$x_{-1} = x_n - f(x_n)/f'(x_n)$. **First locate the root by sign change** — tabulate f at a few integers and find where it flips sign. Start inside that bracket. Iterate to three decimals.

WORKED — M24 2(B) / P26 2(C)

Find a *negative* root of $x\,e_x - \sin x = 0$ to three decimals.

x	−0.5	−1	−2	−2.9	−3.0	−3.5
f(x)	+0.176	+0.474	+0.639	+0.080	−0.008	−0.456

Sign change in $(-3.0,\ -2.9)$. With $f' = e_x + x\,e_x - \cos x$ and $x_0 = -3$:

$x_1 = -2.990745$ → $x_2 = -2.990735$ → converged. **root = −2.991**

TRAP — THIS ONE IS NASTY

$x = 0$ **is itself a root** of this equation. Start anywhere near zero (−0.5, −1) and Newton-Raphson slides straight back to 0, which is not negative — you get 0.000 and lose the question. **You must bracket first.** The first genuine negative root is near −3.

3 Taylor / Maclaurin expansion — two variables 4× Q2(c)

METHOD

$f(x,y) = f(a,b) + [h f_x + k f_y] + (1/2!)[h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}] + \dots$ with $h = x-a$, $k = y-b$. Compute the five partials, evaluate all at (a,b), substitute.

WORKED — M23 2(C): EXPAND $XY + COS(XY)$ ABOUT $(1, \Pi/2)$

At $(1, \pi/2)$ we have $xy = \pi/2$, so $\sin(xy) = 1$ and $\cos(xy) = 0$.

	expression	value at (1, π/2)
f	$xy + \cos(xy)$	$\pi/2$
f_x	$y(1 - \sin xy)$	0
f_y	$x(1 - \sin xy)$	0
f_{xx}	$-y^2 \cos xy$	0
f_{xy}	$1 - \sin xy - xy \cos xy$	0
f_{yy}	$-x^2 \cos xy$	0

Every term vanishes. To second degree, $f(x,y) = \pi/2$.

TRAP — DO NOT PANIC AND DO NOT "FIX" IT

For the $\cos$ version the answer really is just $\pi/2$. Students get zeros, assume they have blundered, and waste ten minutes. **Write the table, state that all first and second derivatives vanish at the point, conclude $f \approx \pi/2$.** Full marks.

The other two variants are NOT trivial — know them:

- $xy + \sin(xy)$: $f = 1+\pi/2$, $f_x = \pi/2$, $f_y = 1$, $f_{xx} = -\pi^2/4$, $f_{xy} = 1-\pi/2$, $f_{yy} = -1$
- $xy^2 + \cos(xy)$: $f = \pi^2/4$, $f_x = \pi(\pi-2)/4$, $f_y = \pi-1$, $f_{xx} = 0$, $f_{xy} = \pi-1$, $f_{yy} = 2$

4 Taylor / Maclaurin expansion — one variable 5× Q1(b)

METHOD

Maclaurin: $f(x) = f(0) + x f'(0) + x^2 f''(0)/2! + \dots$

"In powers of $(x-a)$" means Taylor about a.

Shortcut for inverse-trig and log forms: differentiate once, expand the resulting *rational* function as a binomial series, then integrate term by term. Far faster than repeated differentiation.

WORKED — MAE23 1(C): MACLAURIN SERIES FOR $TAN\ X$ UP TO X^4

$f = \tan x$, $f' = \sec^2 x = 1 + f^2$. Differentiating repeatedly using this identity (much quicker than the quotient rule):

$f' = 1+f^2$, $f'' = 2f f'$, $f''' = 2f'^2 + 2f f''$, $f^{iv} = 6f'f'' + 2f f'''$

At $x=0$: $f=0,\ f'=1,\ f''=0,\ f'''=2,\ f^{iv}=0$

$\tan x = x + x^3/3 + \dots$ (the x^2 and x^4 terms are zero — tan is odd)

TRAP

For an odd function every even-order term vanishes. Say so explicitly rather than leaving gaps — it shows you knew, and it is often worth a mark.

5 Maxima and minima of two variables 5× Q1(d)/Q2(b)

METHOD — THE R-S-T TEST

1. Solve $f_x = 0$, $f_y = 0$ simultaneously for stationary points.
2. Set $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$.

3. At each point: $rt - s^2 > 0$ and $r < 0$ → **maximum**; $rt - s^2 > 0$ and $r > 0$ → **minimum**; $rt - s^2 < 0$ → **saddle**; $= 0$ → test fails.

WORKED — M23 2(B): $X^3 + Y^3 - 3AXY$

$f_x = 3x^2 - 3ay = 0$, $f_y = 3y^2 - 3ax = 0$ → stationary points $(0,0)$ and (a,a).

$r = 6x$, $s = -3a$, $t = 6y$, so $rt - s^2 = 36xy - 9a^2$.

- At $(0,0)$: $rt-s^2 = -9a^2 < 0$ → **saddle point**
- At (a,a): $rt-s^2 = 27a^2 > 0$, $r = 6a$ → **minimum if a > 0, maximum if a < 0**

TRAP

The sign of a flips the conclusion — you must split the case, not assume $a>0$. And $(0,0)$ is a saddle, not "no conclusion": $rt-s^2$ is strictly negative.

6 Lagrange multipliers 4× Q1(d)/Q2(d)

METHOD

To optimise f subject to $g = 0$: form $F = f - \lambda g$ and solve $F_x = F_y = F_\beta = 0$ together with $g = 0$ — four equations, four unknowns. Divide pairs of equations to eliminate λ early; it is almost never wanted.

WORKED A — M23 2(D) / MAM24 1(D): OPEN-TOP BOX, VOLUME 108 FT³, MINIMUM SURFACE AREA

Minimise $S = xy + 2xz + 2yz$ subject to $xyz = 108$.

$F = xy + 2xz + 2yz - \lambda(xyz - 108)$:

$y + 2z = \lambda yz$, $x + 2z = \lambda xz$, $2x + 2y = \lambda xy$

First two give $x = y$; combining with the third gives $x = 2z$. Then $xyz = 108$ → $4z^3 = 108$ → $z = 3$.

x = 6, y = 6, z = 3 ft — minimum surface area S = 108 ft²

WORKED B — M23 1(D): HOTTEST POINT ON THE UNIT SPHERE, $T = 400XYZ^2$

Maximise T on $x^2+y^2+z^2 = 1$. The stationary conditions give $x^2 = y^2 = z^2/2$, so $x^2 = y^2 = 1/4$, $z^2 = 1/2$.

Highest temperature T = 50, at (±½, ±½, ±1/√2) with xy > 0

TRAP

"Open at the top" means **one** xy face, not two — using $2xy$ gives a cube and loses the question. And in the sphere problem z appears squared, so $z^2 = 2x^2$, not $z^2 = x^2$.

UNIT II — Q3 and Q4

Read this first. The four numerical methods share one skeleton and are far faster learned together than apart. Given $y' = f(x,y)$, $y(x_0) = y_0$, step h:
Taylor — differentiate the ODE repeatedly, evaluate at the start point, substitute into the Taylor series. **Euler** — one slope, at the left end. **Modified Euler** — average the slopes at both ends, iterating until the value settles. **RK4** — a weighted average of four slopes. Same tabulation discipline throughout: **carry 4–6 decimals, show every row.**

7 Taylor's series method 5× Q3(c)/Q4(b)

METHOD

$y(x_0+h) = y_0 + h\, y'_0 + (h^2/2!)y''_0 + (h^3/3!)y'''_0 + (h^4/4!)y^{iv}_0 + \dots$
Get y'' by differentiating $y' = f(x,y)$ *totally*: $y'' = f_x + f_y \cdot y'$. Repeat. Then evaluate every derivative at the initial point.

WORKED — MP 3(C) / P26 9(C) VERBATIM REPEAT

$dy/dt = e^{-2t} - 2y$, $y(0) = 0.1$, find $y(0.1)$ to 4th degree.

derivative	expression	value at t=0, y=0.1
y'	$e^{-2t} - 2y$	0.8
y''	$-2e^{-2t} - 2y'$	−3.6
y'''	$4e^{-2t} - 2y''$	11.2
y^{iv}	$-8e^{-2t} - 2y'''$	−30.4

$y(0.1) = 0.1 + (0.1)(0.8) + (0.01/2)(-3.6) + (0.001/6)(11.2) + (0.0001/24)(-30.4)$

$y(0.1) = 0.163740$ (exact $y = (0.1+t)e^{-2t}$ gives 0.163746 — agrees to 5 dp)

TRAP

y'' is a **total** derivative — you must substitute y' back in, not treat y as constant. Read whether the paper says 3rd or 4th degree; MP says 4th, M24 says 3rd.

8 Euler's method 4× Q3(b)/Q4(b)

METHOD

$y_{-1} = y_n + h\ f(x_n,\ y_n)$. One slope per step, taken at the left end. Tabulate.

WORKED — M23 4(B) / MAE23 3(B): BUNGEE JUMPER 2×

$dv/dt = g - (c/m)v^2$, $g = 9.8$, $c = 0.25$, $m = 68.1$, $v(0) = 0$, $h = 1$ s. So $c/m = 0.003671$.

t	v	f = 9.8 − 0.003671v²	v + h·f
0	0	9.8000	9.8000
1	9.8000	9.4475	19.2475
2	19.2475	8.4401	**27.6876**

Velocity after 3 s **≈ 27.69 m/s**. (MAE23 asks only for the *third* second: 8.44 m/s² × 1 s of gain.)

TRAP

Compute c/m once and keep it to 6 significant figures — rounding it to 0.004 visibly changes the third row.

9 Modified Euler's method 5× Q4(c)

METHOD — PREDICTOR / CORRECTOR

Predictor: $y_{-1}^{(0)} = y_n + h\ f(x_n,\ y_n)$
Corrector (iterate): $y_{-1}^{(k+1)} = y_n + (h/2)[f(x_n,y_n) + f(x_{-1},\ y_{-1}^{(k)})]$
Repeat the corrector until two successive values agree — papers usually say "two iterations at each stage".

WORKED — P26 9(D)

$dy/dx = -y/(1+x)$, $y(3) = 2$, find $y(3.2)$, $h = 0.2$.

$f(3,\ 2) = -2/4 = -0.5$

step	value
Predictor	$2 + 0.2(-0.5) = 1.900000$
Corrector 1	$2 + 0.1[-0.5 + (-1.9/4.2)] = 1.904762$
Corrector 2	$2 + 0.1[-0.5 + (-1.904762/4.2)] = 1.904649$
Corrector 3	1.904651 — settled

$y(3.2) \approx 1.9047$ (exact $y = 8/(1+x)$ gives 1.904762)

TRAP

The corrector uses f at the **new** x with the **latest** y estimate, while $f(x_n, y_n)$ stays fixed throughout. Recomputing the first term each time is the standard error.

10 Runge-Kutta, fourth order 6× — every paper Q3(d)/Q4(d)

METHOD

$k_1 = h\,f(x_n, y_n)$ · $k_2 = h\,f(x_n+h/2, y_n+k_1/2)$ · $k_3 = h\,f(x_n+h/2, y_n+k_2/2)$ · $k_4 = h\,f(x_n+h, y_n+k_3)$

$\Delta y = (k_1 + 2k_2 + 2k_3 + k_4)/6$, $y_{-1} = y_n + \Delta y$

WORKED — P26 10(D)

$dy/dx = y - x^2$, $y(2) = 5$, find $y(2.1)$, $h = 0.1$.

k	argument	value
k_1	$0.1 \cdot f(2, 5)$	0.100000
k_2	$0.1 \cdot f(2.05, 5.05)$	0.084750
k_3	$0.1 \cdot f(2.05, 5.042375)$	0.083988
k_4	$0.1 \cdot f(2.1, 5.083988)$	0.067399

$\Delta y = (0.1 + 2(0.08475) + 2(0.083988) + 0.067399)/6 = 0.084146$

$y(2.1) = 5.084146$

TRAP

k_3 uses k_2 (not k_1), and k_4 uses the **full** step h , not $h/2$. The k values already contain h — do not multiply by h again at the end.

11 Newton's law of cooling 5× Q3(b)/(c)/(d), Q4(d)

METHOD

$dT/dt = -k(T - T_0)$ → $T = T_0 + (T_1 - T_0)e^{-kt}$ (T_0 = room temperature, T_1 = initial temperature). Use the one intermediate reading to find k , then solve for the time asked.

WORKED — MP 3(D) / P26 6(C) VERBATIM REPEAT

Coffee 92°C, room 24°C, cools to 80°C in 1 minute. When does it reach 65°C?

$T = 24 + 68\,e^{-kt}$. At $t = 1$: $80 = 24 + 68e^{-k}$ → $e^{-k} = 56/68 = 14/17$ → $k = \ln(17/14) = 0.194156$

For $T = 65$: $41 = 68e^{-kt}$ → $t = \ln(68/41)/0.194156$

t = 2.606 minutes (about 2 min 36 s)

TRAP

Subtract the room temperature *before* taking logs. The M24 variant uses 90 / 24 / 75 / 60 — same working, different numbers.

12 Orthogonal trajectories — Cartesian 6× Q3(b)/(d)

METHOD

1. Differentiate the family and **eliminate the parameter** to get its DE.
2. Replace dy/dx by $-dx/dy$.

3. Solve the new DE.

WORKED — P26 5(C): CO-AXIAL CIRCLES $x^2 + y^2 + 2gx + c = 0$, G THE PARAMETER

Differentiate: $2x + 2y\,y' + 2g = 0$ → $g = -(x + y\,y')$. Substituting back eliminates g:

$x^2 + y^2 - 2x(x + y\,y') + c = 0$ → $y^2 - x^2 + c = 2xy\,y'$

Replace $y' \to -1/y'$: $y^2 - x^2 + c = -2xy/y'$, a homogeneous equation. Solving gives the orthogonal family $\mathbf{x^2 + y^2 + 2\lambda y - c = 0}$ — the conjugate system of co-axial circles.

TRAP

You must eliminate the parameter **before** swapping y'. Swapping first and eliminating after is the commonest way to lose this question.

13 Orthogonal trajectories — polar, and self-orthogonal families 3× Q4(c)

METHOD — POLAR

Eliminate the parameter to get $F(r, \theta, dr/d\theta) = 0$, then replace $dr/d\theta$ by $-r^2(d\theta/dr)$ — equivalently replace $r\,d\theta/dr$ by $-(1/r)(dr/d\theta)$.

WORKED A — MAM24 3(B) / P26 6(B): SHOW $R = B\sin\theta$ IS ORTHOGONAL TO $R = A\cos\theta$

For $r = a\cos\theta$: $dr/d\theta = -a\sin\theta$, and eliminating a gives $(1/r)(dr/d\theta) = -\tan\theta$.

Swap: $(1/r)(dr/d\theta) \to \cot\theta$, i.e. $dr/r = \cot\theta\,d\theta$ → $\ln r = \ln\sin\theta + \ln b$ → $\mathbf{r = b\sin\theta}$ ✓

WORKED B — M23 4(D) / MAE23 4(C): SHOW $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$ IS SELF-ORTHOGONAL

Differentiating and eliminating λ gives a DE in which λ satisfies a quadratic. The key algebraic fact: the resulting differential equation contains y' only through the combination $(x + y\,y')(y - x\,y')$-type symmetric terms, so substituting $y' \to -1/y'$ reproduces the *same* equation.

The family is therefore its own orthogonal trajectory — self-orthogonal. (These are confocal conics: every ellipse of the family cuts every hyperbola of it at right angles.)

TRAP

Do not try to solve for λ explicitly. Show that the DE is *invariant* under $y' \to -1/y'$ — that is the whole proof.

14 LR and RC circuits 3× Q3(d)/Q4(d)

METHOD

LR: $L(di/dt) + Ri = E(t)$ · **RC:** $R(dq/dt) + q/C = E(t)$, with $i = dq/dt$
LC, no e.m.f. (the 2-marker): $L(d^2q/dt^2) + q/C = 0$
All are linear first-order — integrating factor $e^{Rt/L}$ or $e^{t/RC}$.

WORKED — MAE23 3(D)

$e = 200e^{-5t}$ applied to $R = 20\Omega$, $C = 0.01$ F, no initial charge.

$20(dq/dt) + 100q = 200e^{-5t}$ → $dq/dt + 5q = 10e^{-5t}$

Integrating factor e^{5t}: $d(qe^{5t})/dt = 10$ → $qe^{5t} = 10t + A$, and $q(0)=0$ gives $A = 0$.

$\mathbf{q(t) = 10t\,e^{-5t}}$ $\mathbf{i(t) = dq/dt = 10(1 - 5t)e^{-5t}}$

TRAP

Here the forcing term e^{-5t} matches the complementary function, so the particular integral picks up a factor of t — that is why the answer is $10te^{-5t}$ and not a plain exponential. Recognising this resonance case is the whole question. Note $1/C = 100$, not 0.01.

UNIT III — Q5 and Q6

Everything in this unit sits on CF + P.I. Learn that first; Cauchy, Legendre, IVP/BVP and variation of parameters are all built on top of it. Cauchy and Legendre in particular *reduce* to a constant-coefficient problem, so you get them almost free.

15 Complementary function + particular integral 6× Q5(c)/Q6(c)/(d)

CF — FROM THE ROOTS OF THE AUXILIARY EQUATION

Roots	Contribution to CF
real, distinct m_1, m_2	$C_1e^{m_1x} + C_2e^{m_2x}$
real, repeated m (twice)	$(C_1 + C_2x)e^{mx}$
repeated k times	$(C_1 + C_2x + \dots + C^kx^{k-1})e^{mx}$
complex $\alpha \pm i\beta$	$e^{Nx}(C_1\cos\beta x + C_2\sin\beta x)$
complex repeated	$e^{Nx}[(C_1+C_2x)\cos\beta x + (C_3+C_4x)\sin\beta x]$

P.I. — THE STANDARD CASES

RHS	Rule	If it fails
e^{ax}	$e^{ax}/f(a)$	$f(a)=0$ → multiply by x and differentiate f; repeat
$\sin ax$, $\cos ax$	replace D^2 by $-a^2$	denominator 0 → multiply by x, differentiate
x^n	expand $[f(D)]^{-1}$ in ascending powers of D up to D^n	—
$e^{ax}V(x)$	$e^{ax}\cdot[1/f(D+a)]V(x)$ — the shift rule	—
$x\ V(x)$	$[x - f'(D)/f(D)]\cdot[1/f(D)]V$	—

WORKED — P26 5(D): $(D^2 + 2D + 1)Y = X\ E^{2x}$

CF: $m^2+2m+1 = (m+1)^2 = 0$ → $m = -1$ twice → $y^c = (C_1 + C_2x)e^{-x}$

P.I.: shift rule with $a = 2$: $y_\gamma = e^{2x}\cdot[1/((D+2)^2+2(D+2)+1)]x = e^{2x}\cdot[1/(D^2+6D+9)]x$

$1/(9 + 6D + D^2) = (1/9)(1 + (6D+D^2)/9)^{-1} \approx (1/9)(1 - 6D/9 + \dots)$

$y_\gamma = e^{2x}(1/9)(x - 2/3) = e^{2x}(3x-2)/27$

$\mathbf{y = (C_1 + C_2x)e^{-x} + (3x - 2)e^{2x}/27}$

TRAP

Repeated roots need the x factor in the CF. And when using the shift rule, replace **every** D by $D+a$, then expand in ascending powers — descending powers give an infinite series that never terminates.

16 Cauchy's linear differential equation 4× Q5(c)/(d)

METHOD — THE 4 STEPS (THIS IS ALSO THE 2-MARK ANSWER)

1. Put $x = e^z$, i.e. $z = \log x$. Write $D = d/dz$.
2. Then $x\,y' = Dy$, $x^2y'' = D(D-1)y$, $x^3y''' = D(D-1)(D-2)y$.
3. The equation becomes constant-coefficient in z. Solve for CF and P.I. as usual.
4. Substitute back $z = \log x$.

WORKED — MAE23 6(D): $X^2Y'' - 4XY' + 6Y = \cos(2\log X)$

With $x = e^z$: $D(D-1)y - 4Dy + 6y = \cos 2z$ → $(D^2 - 5D + 6)y = \cos 2z$

CF: $m^2-5m+6 = (m-2)(m-3)$ → $C_1e^{2z} + C_2e^{3z} = C_1x^2 + C_2x^3$

P.I.: replace $D^2 \to -4$: $1/(-4-5D+6)\cos 2z = 1/(2-5D)\cos 2z$

Multiply above and below by $(2+5D)$: $(2+5D)\cos 2z/(4-25D^2) = (2\cos 2z - 10\sin 2z)/(4+100)$

$$y = C_1x^2 + C_2x^3 + [\cos(2 \log x) - 5 \sin(2 \log x)]/52$$

TRAP

$x^2y'' = D(D-1)y$, **not** D^2y — the $-D$ is the single most common slip in this unit. Also $\cos(2 \log x)$ becomes $\cos 2z$, so treat z as the variable throughout and only convert back at the very end.

17 Legendre's linear differential equation 4× Q6(c)/(d)

METHOD — IDENTICAL TO CAUCHY, ONE EXTRA FACTOR

Put $ax + b = e^z$. Then
$(ax+b)y' = aDy$ · $(ax+b)^2y'' = a^2D(D-1)y$ · $(ax+b)^3y''' = a^3D(D-1)(D-2)y$
Everything else is the same. **Cauchy is just Legendre with a=1, b=0.**

WORKED — MAE23 5(B): $(2X+3)^2Y'' + 2(2X+3)Y' - 12Y = 0$

Here $a = 2$. Put $2x+3 = e^z$:

$4D(D-1)y + 2\cdot 2Dy - 12y = 0$ → $4D^2 - 4D + 4D - 12 = 0$ → $4D^2 = 12$ → $m = \pm\sqrt{3}$

$y = C_1e^{\sqrt{3}z} + C_2e^{-\sqrt{3}z}$

$$y = C_1(2x+3)^{\sqrt{3}} + C_2(2x+3)^{-\sqrt{3}}$$ *(verified by substitution)*

TRAP

Forgetting the a^2 on the second-derivative term. With $a = 2$ that factor is 4 — drop it and the auxiliary equation, and every root, is wrong.

18 Method of variation of parameters 5× — every paper Q5(c)/(d)

METHOD

For $y'' + Py' + Qy = R$ with CF $y_c = C_1y_1 + C_2y_2$:
1. Wronskian $W = y_1y_2' - y_2y_1'$
2. $y_p = -y_1\int(y_2R/W)dx + y_2\int(y_1R/W)dx$

The equation must be in standard form — coefficient of y'' equal to 1 — before you read off R.

WORKED — MAM24 6(C): $Y'' - 6Y' + 9Y = E^{3x}/X^2$

CF: $(m-3)^2 = 0$ → $y_1 = e^{3x}$, $y_2 = xe^{3x}$

Wronskian: $W = e^{3x}(e^{3x} + 3xe^{3x}) - xe^{3x}\cdot 3e^{3x} = e^{6x}$

$y_2R/W = xe^{3x}\cdot(e^{3x}/x^2)/e^{6x} = 1/x$ → $\int = \log x$

$y_1R/W = e^{3x}\cdot(e^{3x}/x^2)/e^{6x} = 1/x^2$ → $\int = -1/x$

$y_p = -e^{3x}\log x + xe^{3x}(-1/x) = -e^{3x}(\log x + 1)$

$$y = (C_1 + C_2x - \log x)e^{3x}$$ *(the −1 is absorbed into C_1)*

TRAP

Papers choose an R that is deliberately awkward for the P.I. rules — $\sec^3x$, $\tan x$, e^{3x}/x^2, $\cos(e^{-x})$. That awkwardness is *the signal* to use variation of parameters. Watch the minus sign on the first integral.

19 Initial and boundary value problems 6× Q5(b)/Q6(b)/(d)

METHOD

Find the general solution first (CF + P.I.), **then** apply the conditions. **IVP** = both conditions at the same x (e.g. $y(0)$ and $y'(0)$). **BVP** = conditions at two different x values.

WORKED A — P26 5(B), BVP: $Y'' + Y = 0$, $Y(0)=2$, $Y(\Pi/2) = -2$

$y = A \cos x + B \sin x$. $y(0) = A = 2$; $y(\pi/2) = B = -2$.

$y = 2\cos x - 2\sin x = 2\sqrt{2} \cos(x + \pi/4)$

WORKED B — M23 6(D), IVP: $(D^2 - 2D)Y = E_x \text{ SIN } X$, $Y(0)=8$, $Y'(0)=1$

CF: $m(m-2)=0$ → $C_1 + C_2e^2{}_x$

P.I.: shift rule, $e_x \cdot [1/((D+1)^2-2(D+1))]\sin x = e_x[1/(D^2-1)]\sin x$; replace $D^2 \to -1$: $= -e_x\sin x/2$

General: $y = C_1 + C_2e^2{}_x - e_x\sin x/2$. Applying both conditions:

$y = 29/4 + (3/4)e^2{}_x - e_x\sin x/2$

TRAP

Apply conditions to the **complete** solution including the P.I. — fitting constants to the CF alone is a guaranteed zero. Differentiate the full expression before using $y'(0)$.

UNIT IV — Q7 and Q8

Build the difference table first, every time. Interpolation, missing terms, numerical differentiation and grouped-frequency questions all start from the same table — only the formula you feed it into changes. Draw it once, carefully, and half the unit is done.

20 Newton-Gregory forward and backward interpolation 2× Q7(c)

METHOD

Forward (use when x is near the *start*), $p = (x - x_0)/h$:

$y = y_0 + p\Delta y_0 + p(p-1)/2! \cdot\Delta^2 y_0 + p(p-1)(p-2)/3! \cdot\Delta^3 y_0 + \dots$

Backward (use when x is near the *end*), $p = (x - x_n)/h$:

$y = y_n + p\nabla y_n + p(p+1)/2! \cdot\nabla^2 y_n + p(p+1)(p+2)/3! \cdot\nabla^3 y_n + \dots$

Note the sign pattern: forward has $(p-1)(p-2)$, backward has $(p+1)(p+2)$.

WORKED — MP 7(C): COMPUTE $F(14.2)$

x	10	12	14	16	18
y	0.240	0.281	0.318	0.352	0.384

Difference table (h = 2):

Δ	0.041	0.037	0.034	0.032
Δ²	−0.004	−0.003	−0.002	
Δ³	0.001	0.001		
Δ⁴	0.000			

Backward from $x_n = 18$: $p = (14.2 - 18)/2 = -1.9$. Backward differences at the end: $\nabla y = 0.032$, $\nabla^2 y = -0.002$, $\nabla^3 y = 0.001$.

f(14.2) = 0.321518

TRAP

Choosing the wrong end. $x = 14.2$ sits past the middle, so backward is correct here; using forward from 10 needs many more terms and drifts. **Pick the end nearest your x.**

21 Lagrange's interpolation (and inverse) 3× Q7(b)/(c), Q8(c)

METHOD

$y = \Sigma_i\ y_i \cdot \Pi_{j\neq i}\ (x - x_j)/(x_i - x_j)$

Inverse interpolation — to find x given y — is the identical formula with the roles of x and y swapped: $x = \Sigma_i\ x_i \cdot \Pi_{j\neq i}\ (y - y_j)/(y_i - y_j)$

Works for **unequally spaced** data — that is when to reach for it.

WORKED — MAM24 7(B): FIND F(1) FROM (−1, 9), (0, 5), (2, 3)

$y = 9\cdot(x-0)(x-2)/((-1)(-3)) + 5\cdot(x+1)(x-2)/((1)(-2)) + 3\cdot(x+1)(x-0)/((3)(2))$

Simplifying: $P(x) = x^2 - 3x + 5$

f(1) = 1 − 3 + 5 = 3

TRAP

The denominators use the x_i values, never x. For the 2-mark "write Lagrange's inverse interpolation formula" for two points, the answer is simply $x = x_0(y-y_1)/(y_0-y_1) + x_1(y-y_0)/(y_1-y_0)$.

22 Newton's divided difference 5× — every paper Q7(d)/Q8(c)

METHOD

$f[x_0,x_1] = (f_1-f_0)/(x_1-x_0)$, and higher orders recursively.

$P(x) = f_0 + (x-x_0)f[x_0,x_1] + (x-x_0)(x-x_1)f[x_0,x_1,x_2] + \dots$

Also works for unequal spacing, and unlike Lagrange it is easy to extend when a point is added.

WORKED — MP 8(C) / M24 7(D) VERBATIM REPEAT

Cubic through $(2,4), (4,56), (9,711), (10,980)$; estimate y at $x = 1.5$.

x	f	1st	2nd	3rd
2	4	26	15	1
4	56	131	23	
9	711	269		
10	980			

$P(x) = 4 + 26(x-2) + 15(x-2)(x-4) + 1\cdot(x-2)(x-4)(x-9)$

Expanding: **$P(x) = x^3 - 2x$**

y(1.5) = 3.375 − 3 = 0.375

TRAP

The polynomial collapses to the very clean $x^3 - 2x$ — if your expansion has stray x^2 or constant terms, you have an arithmetic slip. Expand fully: the question asks for the polynomial, not just the value.

23 Finding missing terms 3× Q8(b)

METHOD

With n known values, a polynomial of degree $n-1$ fits, so all differences of order n vanish. Set $\Delta^n y_i = 0$ for as many i as you have unknowns, and solve.

3 known values → quadratic fits → set every third difference to zero.

WORKED — MAM24 8(B) / P26 8(B) IDENTICAL DATA TWICE

x	45	50	55	60	65
y	3.0	**a**	2.0	**b**	−2.4

Three known values → quadratic → both third differences vanish:

$\Delta^3 y_0 = 3a + b - 9 = 0$ and $\Delta^3 y_1 = -a - 3b + 3.6 = 0$

Solving: **y(50) = a = 2.925** **y(60) = b = 0.225**

Check — second differences become $-0.85, -0.85, -0.85$, constant as required. Always show this check; it is worth a mark.

TRAP

Two unknowns need **two** equations. Setting a single fourth difference to zero gives only the relation $a + b = 3.15$ and cannot be solved — students get stuck here. Use the *third* differences, both of them.

24 Grouped-frequency estimation 3× Q7(c)/Q8(c)/(d)

METHOD — THE PART NOBODY SPOTS

The data are given *per class*, but interpolation needs a **cumulative** ("less than") column.

1. Convert the class frequencies to cumulative totals against the class *upper bounds*.
2. Interpolate the cumulative value at the point asked.
3. **Subtract** to recover the count in the sub-interval.

WORKED — P26 7(C): STUDENTS SCORING BETWEEN 70 AND 75

Marks	30–40	40–50	50–60	60–70	70–80
Students	31	42	51	35	31

Cumulative — students scoring *less than*:

x (marks <)	40	50	60	70	80
y	31	73	124	159	190

Forward differences: $\Delta = 42, 51, 35, 31$; $\Delta^2 = 9, -16, -4$; $\Delta^3 = -25, 12$; $\Delta^4 = 37$.

$p = (75 - 40)/10 = 3.5$, giving $y(75) = 172.80$.

Students between 70 and 75 = 172.80 − 159 = 13.80 ≈ 14

TRAP

Interpolating the raw class frequencies instead of the cumulative column. That is the whole difficulty of this question — and it is worth 7 marks three papers running.

25 Numerical differentiation 4× Q7(d)/Q8(c)

METHOD

Differentiate the interpolation polynomial. **Forward**, with $p = (x-x_0)/h$:

$dy/dx = (1/h)[\Delta y_0 + (2p-1)/2 \cdot\Delta^2 y_0 + (3p^2-6p+2)/6 \cdot\Delta^3 y_0 + \ldots]$

Backward, with $p = (x-x_n)/h$:

$dy/dx = (1/h)[\nabla y_n + (2p+1)/2 \cdot\nabla^2 y_n + (3p^2+6p+2)/6 \cdot\nabla^3 y_n + \ldots]$

At $x = x_0$ exactly, $p = 0$ and the forward formula collapses to $(1/h)[\Delta y_0 - \Delta^2 y_0/2 + \Delta^3 y_0/3 - \ldots]$.

WORKED — P26 7(D): RATE OF COOLING AT $T = 8$ S

t	1	3	5	7	9
θ	85.3	74.5	67	60.5	54.3

$h = 2$. Since $t = 8$ is near the end, use backward from $t_n = 9$, $p = (8-9)/2 = -0.5$.

Backward differences at the end: $\nabla\theta = -6.2$, $\nabla^2\theta = 0.3$, $\nabla^3\theta = -0.7$, $\nabla^4\theta = 1.6$.

dθ/dt at t = 8 is −3.1187 °C per second

TRAP

Do not forget the $1/h$ outside — with $h = 2$ that halves your answer. The result is negative because the body is cooling; a positive answer means a sign slip.

26–28 Numerical integration — all three rules 5× Q7(d)/Q8(b)/(d)

THE THREE FORMULAS

Trapezoidal (any n): $(h/2)[(y_0+y_n) + 2(\text{all interior})]$
Simpson's 1/3 (n even): $(h/3)[(y_0+y_n) + 4(\text{odd-indexed}) + 2(\text{even-indexed interior})]$
Simpson's 3/8 (n a multiple of 3): $(3h/8)[(y_0+y_n) + 3(\text{all not divisible by 3}) + 2(\text{indices divisible by 3})]$
"Seven equidistant ordinates" means $n = 6$ intervals.

WORKED — THE RECURRING INTEGRAL $\int_0^1 dx/(1+x^2)$ **3 PAPERS, 3 DIFFERENT RULES**

Exact value $= \pi/4 = 0.785398$. Ordinates for $n = 6$, $h = 1/6$:

x	0	1/6	2/6	3/6	4/6	5/6	1
y	1.000000	0.972973	0.900000	0.800000	0.692308	0.590164	0.500000

Rule	Result	×4 → π	Paper
Trapezoidal, 4 strips	**0.782794**	3.131176	MP 8(b)
Simpson's 1/3, 6 intervals	**0.785398**	3.141592	M23 7(c)
Simpson's 3/8, 6 intervals	**0.785396**	3.141583	MAM24 8(d)

Simpson's 1/3 recovers π to six decimal places — which is exactly why the paper keeps asking it this way.

TRAP

Simpson's 1/3 needs an **even** number of intervals; 3/8 needs a **multiple of 3**. "Seven ordinates" = 6 intervals, which satisfies both — that is why it is the paper's favourite. Count ordinates from y_0, and never apply the 4-2-4-2 weights to 3/8.

UNIT V — Q9 and Q10

Eigenvalues are the spine of this unit. Diagonalization, Rayleigh's power method and system-of-ODEs-by-matrix are all "find the eigenvalues, then do one more step". Learn the eigen-step once and three 7-markers open at the same time.

29 Rank, echelon form and consistency 2× Q9(b)/Q10(c)

METHOD

Row-reduce the augmented matrix $[A|B]$. With n unknowns:

- $\text{rank}(A) \neq \text{rank}(A|B)$ → **inconsistent**, no solution
- $\text{rank}(A) = \text{rank}(A|B) = n$ → **unique** solution
- $\text{rank}(A) = \text{rank}(A|B) = r < n$ → **infinitely many**, with $n - r$ free parameters

WORKED — CONDITIONS ON A, B, C FOR SOLVABILITY

$x + 2y - z = a$; $2x + y + z = b$; $x - y + 2z = c$

The coefficient determinant is **0** and `rank(A) = 2`, so a condition is needed. Adding rows 1 and 3: `2x + y + z = a + c`. But row 2 says `2x + y + z = b`.

Condition for consistency: b = a + c

When it holds, `rank = 2 < 3`, so there is a one-parameter family. Taking `a=1, b=3, c=2` and `z = t` free:

x = 5/3 − t,   y = t − 1/3,   z = t

TRAP — AND A NOTE ON THE 2023 PAPER

When `rank < n` you must present a *family* with a named parameter, not a single triple.
The exact constants in M23 10(c) are not legible in the scan — the method above is what matters, and the structure (det = 0, one linear condition on a, b, c, one free parameter) is standard.

30 Gauss elimination 2× Q9(d)/Q10(c)

METHOD

Form `[A|B]`, reduce to upper triangular by row operations, then back-substitute. In word problems the real work is **writing the system** — one equation per constraint, one unknown per quantity asked for.

WORKED — M24 10(C): THE DIETICIAN PROBLEM

	Food 1	Food 2	Food 3	Required
Vitamin A	10	30	20	200
Vitamin C	50	30	25	250
Calcium	60	20	40	300

With `x, y, z` grams of Foods 1, 2, 3:

`10x + 30y + 20z = 200`; `50x + 30y + 25z = 250`; `60x + 20y + 40z = 300`

x = 5/8 = 0.625 g,   y = 25/8 = 3.125 g,   z = 5 g

TRAP

Read the table by *row* (one nutrient, all three foods) — transposing it gives a different system and a wrong answer. Divide row 1 by 10 first; the arithmetic gets much easier.

31 Gauss-Seidel iteration 4× Q9(b)/Q10(b)/(c)

METHOD

1. **Reorder the equations for diagonal dominance** — each diagonal entry larger in magnitude than the sum of the others in its row. Without this the iteration may not converge.
2. Solve each equation for its diagonal variable.
3. Sweep in order, **using each new value immediately** within the same iteration. (That is the difference from Jacobi.)

WORKED — MP 10(C) / P26 2(B) IDENTICAL SYSTEM TWICE

Given as: `x + 8y + 2z = −4`; `10x + 2y + z = 59`; `2x − y + 20z = 74`

Reorder to `10x + 2y + z = 59`; `x + 8y + 2z = −4`; `2x − y + 20z = 74`, then

`x = (59 − 2y − z)/10`, `y = (−4 − x − 2z)/8`, `z = (74 − 2x + y)/20`

Iteration	x	y	z
0	0	0	0
1	5.900000	−1.237500	3.048125
2	5.842688	−1.992367	3.016113
3	5.996862	−2.003636	3.000132

4	6.000714	−2.000122	2.999922

Converging to the exact solution **$x = 6, y = -2, z = 3$**

TRAP — THE REORDERING IS THE QUESTION

As printed, this system is not diagonally dominant (row 1 has diagonal 1 against 8 and 2). You must reorder before iterating. Students who charge straight in get numbers that wander and lose most of the marks. Also: use the just-computed `x` when finding `y` in the *same* sweep — holding old values is Jacobi's method, not Gauss-Seidel.

32 Eigenvalues and eigenvectors `foundation` `Q9(c)/(d)`

METHOD

1. Solve $|A - \lambda I| = 0$ for the eigenvalues.
2. For each λ, solve $(A - \lambda I)X = 0$ for the eigenvector.

Checks worth doing: sum of eigenvalues = trace; product = determinant. Both are free marks and catch algebra slips.

WORKED — $A = [[1, -1], [2, 4]]$

$|A - \lambda I| = (1-\lambda)(4-\lambda) + 2 = \lambda^2 - 5\lambda + 6 = (\lambda-2)(\lambda-3)$

$\lambda = 2, 3$. Check: trace $= 5 = 2+3$ ✓, det $= 6 = 2\times3$ ✓

$\lambda=2$: $(A-2I)X=0$ → $-x - y = 0$ → **$X_1 = [-1, 1]^T$**
$\lambda=3$: $-2x - y = 0$ → **$X_2 = [-1, 2]^T$**

TRAP

Eigenvectors are determined only up to a scale factor — any non-zero multiple is correct. Never "solve" to a unique vector and never report the zero vector.

33 Diagonalization and A^n `3×/3` `Q9(c)/(d)`

METHOD

Build the modal matrix P from the eigenvectors as columns, and the spectral matrix $D = \mathrm{diag}(\lambda_i)$ **in the same order**. Then $P^{-1}AP = D$, so $A = PDP^{-1}$ and **$A^n = P\, D^n\, P^{-1}$** — and D^n is just the diagonal entries raised to the power.

WORKED — MP 9(C): DIAGONALIZE $A = [[1,-1],[2,4]]$, HENCE FIND A^8

From above, $P = [[-1, -1], [1, 2]]$ and $D = \mathrm{diag}(2, 3)$.

$P^{-1} = [[-2, -1], [1, 1]]$ (det $P = -1$)

$A^8 = P\cdot\mathrm{diag}(2^8, 3^8)\cdot P^{-1} = P\cdot\mathrm{diag}(256, 6561)\cdot P^{-1}$

$A^8 = [[-6049, -6305], [12610, 12866]]$ *(verified against direct multiplication)*

TRAP

The columns of P and the diagonal of D must be in the **same order**. Swap one and everything after is wrong. For the 2-marker: the *modal* matrix is P, the *spectral* matrix is D. And "is every square matrix diagonalizable?" — **no**: it needs n linearly independent eigenvectors, which repeated eigenvalues can fail to provide.

34 Rayleigh's power method `3×/3` `Q9(c)/(d), Q10(b)`

METHOD

Repeat: $Y = A\,X$, then factor out the largest-magnitude component m so that $X_{-1} = Y/m$. The m values converge to the **dominant** eigenvalue and X to its eigenvector.

WORKED — MP 9(D): $A = [[2,-1,0],[-1,2,-1],[0,-1,2]]$, $X_0 = [1,1,1]^T$

Iter	λ estimate	X
1	1.00000	[1, 0, 1]
2	2.00000	[1, −1, 1]
3	−4.00000	[−0.75, 1, −0.75]
4	3.50000	[−0.71429, 1, −0.71429]
9	3.41423	[−0.70711, 1, −0.70711]
12	3.41421	[−0.70711, 1, −0.70711]

Dominant eigenvalue = $2 + \sqrt{2} = 3.41421$, eigenvector $[-1, \sqrt{2}, -1]^T$

TRAP — THIS SPECIFIC QUESTION IS A TRAP

The start vector $[1,1,1]^T$ is nearly orthogonal to the dominant eigenvector, so the early iterates are wild: 1, then 2, then -4. A student told to "carry out two iterations" would report $\lambda = 2$ — which is a genuine eigenvalue of this matrix but *not* the dominant one. The model paper asks for two decimal places, which needs about **ten** iterations. If the sequence is still jumping, keep going and say so.

35 System of ODEs by matrix method 3×/3 — always Q10(d) Q10(d)

METHOD

Write the system as $X' = AX$.

1. Find the eigenvalues λ_i and eigenvectors V_i of A.
2. General solution $X = c_1 V_1 e^{E_1 t} + c_2 V_2 e^{E_2 t}$.
3. Apply the initial conditions to fix c_1, c_2.

WORKED — M23 10(D): RABBITS AND WOLVES

$dr/dt = 4r - 2w$, $dw/dt = r + w$, $r(0) = 240$, $w(0) = 300$.

$A = [[4, -2], [1, 1]]$, $|A-\lambda I| = \lambda^2 - 5\lambda + 6 = (\lambda-2)(\lambda-3)$

$\lambda = 2 \rightarrow V = [1, 1]^T$; $\lambda = 3 \rightarrow V = [2, 1]^T$

$r = c_1 e^{2t} + 2c_2 e^{3t}$, $w = c_1 e^{2t} + c_2 e^{3t}$

At $t=0$: $c_1 + 2c_2 = 240$, $c_1 + c_2 = 300 \rightarrow c_2 = -60$, $c_1 = 360$

$r(t) = 360e^{2t} - 120e^{3t}$ **$w(t) = 360e^{2t} - 60e^{3t}$**

Check: $r(0) = 360-120 = 240$ ✓, $w(0) = 360-60 = 300$ ✓

TRAP

The eigenvector components must stay attached to the right variable — if $V = [2,1]^T$ belongs to $\lambda=3$, then r gets the 2 and w gets the 1. Always substitute $t = 0$ back at the end; it catches a swapped pair instantly. **This has been the final question of the exam in all three papers.**

Two habits worth more than any single formula.
1. Check your answer when the question hands you a way to. Initial conditions, trace and determinant, constant differences in a table, the exact value of an integral — each is a free verification and often a mark in itself.
2. Show the table. Every iterative method here is method-marked. A Gauss-Seidel, RK4 or difference table with a slip in the last row still scores most of the marks; a bare final answer scores few.

All 35 problem types carrying 7-mark parts, across both options in every unit. Every numerical result above was computed and verified symbolically or numerically before printing; worked examples are taken from the model paper and the 2023, 2024 and 2026 papers wherever one exists. Where a scanned question was not legible, that is stated rather than guessed. Repeat counts are out of the papers carrying each unit (I–III: 6 · IV: 5 · V: 3).